DISCO-TAB — Comprehensive Benchmark

all generators × 24 datasets × all metrics; mean over up to 3 seeds; **bold** = best per row (where a direction is defined); – = missing run/metric.
Generators: DISCO-TAB, CTGAN, TVAE, GaussCop, TabDDPM, ADS-GAN, PATE-GAN, RTVAE, NFlow, ARF, BayesNet, Marginals.

Utility (downstream TSTR)

Table 1: TSTR F1-macro (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.783**	0.483	0.135	0.591	0.406	0.635	0.396	0.600	0.559	0.649	0.735	0.394
breast_cancer	0.930	0.295	0.934	0.912	0.924	0.934	0.497	0.937	0.855	0.940	**0.947**	0.513
liver_disorders	0.509	0.473	0.527	0.553	0.592	0.542	0.488	0.490	0.500	**0.593**	0.561	0.380
parkinsons	0.646	0.480	0.000	0.630	**0.928**	0.737	0.424	0.704	0.695	0.688	0.759	0.446
obesity	**0.886**	0.138	0.490	0.243	0.795	0.655	0.115	0.607	0.171	0.757	0.821	0.102
diabetes	0.843	0.836	**0.875**	0.485	–	0.797	0.478	0.727	0.658	0.853	0.704	0.377
german_credit	**0.646**	0.448	0.137	0.496	–	0.594	0.468	0.501	0.449	0.584	0.638	0.459
bank_marketing	**0.673**	0.604	0.673	0.469	–	0.608	0.469	0.498	0.471	0.558	0.551	0.236
census_income	0.765	0.727	0.746	0.459	–	0.669	0.433	0.549	0.615	0.744	**0.767**	0.524
surv_melanoma	0.536	–	–	–	–	0.557	0.442	0.529	0.508	**0.573**	0.570	0.445
surv_pbc	0.522	–	–	–	–	0.590	0.492	0.646	0.566	0.634	**0.679**	0.428
surv_pbc3	0.468	–	–	–	–	0.632	0.453	0.469	0.477	0.553	**0.707**	0.497
surv_retinopathy	0.518	–	–	–	–	0.539	0.480	0.478	0.513	**0.586**	0.493	0.511
surv_whas500	**0.739**	–	–	–	–	0.699	0.395	0.619	0.709	0.671	0.673	0.324
surv_gbsg2	0.580	–	–	–	–	0.538	0.474	0.556	**0.590**	0.572	0.547	0.475
surv_colon	**0.614**	–	–	–	–	0.497	0.426	0.564	0.526	0.581	0.539	0.450
surv_grace	**0.609**	–	–	–	–	0.591	0.456	0.508	0.543	0.593	0.521	0.463
surv_trace	0.676	–	–	–	–	0.646	0.445	0.516	0.609	**0.704**	0.684	0.516
surv_framingham	**0.581**	–	–	–	–	0.546	0.474	0.486	0.453	0.532	0.475	0.448
tab_blood_transfusion	0.351	–	–	–	–	0.498	0.510	0.517	0.456	**0.580**	0.528	0.502
tab_credit_g	0.643	–	–	–	–	0.577	0.435	0.465	0.448	0.526	**0.653**	0.434
tab_qsar_biodeg	0.796	–	–	–	–	0.729	0.428	0.770	0.612	0.780	**0.850**	0.393
tab_maternal_risk	**0.624**	–	–	–	–	0.463	0.235	0.527	0.395	0.573	0.214	0.234
tab_website_phishing	**0.614**	–	–	–	–	0.555	0.263	0.492	0.454	0.562	0.551	0.286

Table 2: TSTR F1-weighted (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.799**	0.559	0.185	0.679	0.555	0.710	0.500	0.676	0.655	0.722	0.768	0.416
breast_cancer	0.935	0.284	0.939	0.920	0.929	0.939	0.489	0.942	0.869	0.944	**0.950**	0.558
liver_disorders	0.511	0.472	0.529	0.553	0.592	0.543	0.488	0.490	0.501	**0.593**	0.562	0.384
parkinsons	0.732	0.632	0.000	0.729	**0.947**	0.801	0.630	0.784	0.768	0.756	0.829	0.584
obesity	**0.888**	0.138	0.493	0.250	0.796	0.655	0.120	0.610	0.176	0.759	0.827	0.101
diabetes	0.956	0.952	**0.962**	0.876	–	0.945	0.874	0.927	0.901	0.959	0.923	0.584
german_credit	0.668	0.585	0.192	0.624	–	0.680	0.601	0.631	0.587	0.687	**0.715**	0.516
bank_marketing	**0.865**	0.846	0.854	0.828	–	0.862	0.828	0.835	0.829	0.852	0.848	0.273
census_income	0.830	0.812	0.818	0.671	–	0.782	0.656	0.718	0.748	0.824	**0.838**	0.672
surv_melanoma	0.561	–	–	–	–	0.676	0.588	0.645	0.614	**0.684**	0.672	0.454
surv_pbc	0.507	–	–	–	–	0.592	0.528	0.662	0.586	0.648	**0.692**	0.444
surv_pbc3	0.541	–	–	–	–	0.786	0.751	0.749	0.748	0.791	**0.845**	0.675
surv_retinopathy	0.542	–	–	–	–	0.565	0.481	0.507	0.531	**0.601**	0.513	0.517
surv_whas500	**0.752**	–	–	–	–	0.719	0.428	0.642	0.721	0.697	0.696	0.339
surv_gbsg2	0.587	–	–	–	–	0.542	0.486	0.564	**0.596**	0.582	0.564	0.474
surv_colon	**0.614**	–	–	–	–	0.495	0.429	0.565	0.526	0.582	0.540	0.451
surv_grace	0.639	–	–	–	–	**0.656**	0.526	0.591	0.634	0.652	0.617	0.503
surv_trace	0.676	–	–	–	–	0.645	0.444	0.516	0.609	**0.704**	0.684	0.516
surv_framingham	0.622	–	–	–	–	**0.635**	0.590	0.580	0.580	0.628	0.595	0.490
tab_blood_transfusion	0.417	–	–	–	–	0.607	0.636	0.615	0.590	**0.678**	0.652	0.551
tab_credit_g	0.675	–	–	–	–	0.657	0.582	0.606	0.588	0.645	**0.725**	0.501
tab_qsar_biodeg	0.819	–	–	–	–	0.767	0.465	0.799	0.674	0.809	**0.870**	0.361

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
tab_maternal_risk	**0.660**	–	–	–	–	0.510	0.306	0.592	0.485	0.637	0.191	0.251
tab_website_phishing	**0.746**	–	–	–	–	0.694	0.346	0.662	0.614	0.734	0.727	0.368

Table 3: TSTR accuracy (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.794**	0.572	0.228	0.733	0.683	0.750	0.572	0.711	0.717	0.761	0.767	0.472
breast_cancer	0.936	0.316	0.939	0.921	0.930	0.939	0.561	0.942	0.874	0.944	**0.950**	0.579
liver_disorders	0.546	0.502	0.551	0.556	**0.594**	0.551	0.493	0.493	0.507	**0.594**	0.565	0.473
parkinsons	0.735	0.675	0.000	0.744	**0.949**	0.803	0.735	0.795	0.769	0.752	0.846	0.607
obesity	**0.887**	0.154	0.499	0.297	0.801	0.655	0.149	0.627	0.270	0.766	0.828	0.115
diabetes	0.961	0.954	**0.963**	0.915	–	0.953	0.915	0.941	0.909	0.963	0.938	0.479
german_credit	0.655	0.663	0.233	0.700	–	0.712	0.680	0.715	0.668	**0.743**	0.742	0.545
bank_marketing	0.867	0.858	0.845	0.883	–	0.889	0.883	0.884	0.883	**0.891**	0.886	0.266
census_income	0.832	0.823	0.821	0.766	–	0.812	0.758	0.782	0.784	0.836	**0.846**	0.698
surv_melanoma	0.545	–	–	–	–	**0.707**	0.626	0.667	0.634	**0.707**	0.683	0.463
surv_pbc	0.566	–	–	–	–	0.593	0.561	0.667	0.603	0.651	**0.693**	0.439
surv_pbc3	0.543	–	–	–	–	0.781	0.829	0.810	0.805	0.848	**0.857**	0.652
surv_retinopathy	0.554	–	–	–	–	0.581	0.514	0.527	0.532	**0.608**	0.518	0.514
surv_whas500	**0.753**	–	–	–	–	0.728	0.452	0.649	0.720	0.713	0.706	0.333
surv_gbsg2	0.589	–	–	–	–	0.560	0.493	0.565	**0.597**	0.587	0.587	0.488
surv_colon	**0.616**	–	–	–	–	0.550	0.492	0.569	0.535	0.582	0.542	0.475
surv_grace	0.630	–	–	–	–	0.678	0.545	0.625	**0.692**	0.663	0.682	0.495
surv_trace	0.679	–	–	–	–	0.660	0.501	0.518	0.611	**0.704**	0.684	0.519
surv_framingham	0.616	–	–	–	–	0.674	0.660	0.613	0.668	0.674	**0.676**	0.472
tab_blood_transfusion	0.427	–	–	–	–	0.645	0.682	0.623	0.629	**0.701**	0.698	0.542
tab_credit_g	0.663	–	–	–	–	0.683	0.678	0.700	0.668	0.712	**0.747**	0.540
tab_qsar_biodeg	0.820	–	–	–	–	0.779	0.485	0.806	0.709	0.815	**0.874**	0.423
tab_maternal_risk	0.656	–	–	–	–	0.524	0.381	0.626	0.557	**0.678**	0.278	0.289
tab_website_phishing	**0.761**	–	–	–	–	0.708	0.434	0.683	0.637	0.749	0.747	0.336

Statistical fidelity

Table 4: Jensen–Shannon div. (↓)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.054	0.094	0.200	0.042	0.194	0.070	0.192	0.265	0.040	0.041	**0.034**	0.189
breast_cancer	0.062	0.143	0.111	0.027	0.294	0.072	0.568	0.466	0.024	0.016	**0.013**	0.293
liver_disorders	0.068	0.184	0.129	0.028	0.028	0.113	0.118	0.450	0.040	**0.026**	0.029	0.316
parkinsons	0.082	0.175	0.136	0.056	0.227	0.120	0.143	0.578	0.049	0.053	**0.039**	0.269
obesity	0.022	0.044	0.090	0.057	0.044	0.047	0.126	0.158	0.033	0.030	**0.010**	0.200
diabetes	0.024	0.030	0.035	0.067	–	0.062	0.174	0.094	0.032	0.039	**0.007**	0.225
german_credit	0.074	0.020	0.171	0.008	–	0.019	0.046	0.067	0.023	0.007	**0.005**	0.140
bank_marketing	0.016	**0.009**	0.040	0.015	–	0.040	0.088	0.083	0.040	0.014	0.014	0.275
census_income	0.050	**0.019**	0.055	0.049	–	0.082	0.182	0.145	0.085	0.037	0.052	0.347
surv_melanoma	0.064	–	–	–	–	0.077	0.187	0.331	0.033	0.027	**0.023**	0.127
surv_pbc	0.054	–	–	–	–	0.021	0.053	0.093	0.014	0.005	**0.004**	0.070
surv_pbc3	0.045	–	–	–	–	0.051	0.182	0.303	0.031	0.023	**0.016**	0.179
surv_retinopathy	0.126	–	–	–	–	0.025	0.043	0.092	0.020	0.005	**0.004**	0.032
surv_whas500	0.034	–	–	–	–	0.030	0.129	0.244	0.015	0.010	**0.008**	0.129
surv_gbsg2	0.047	–	–	–	–	0.081	0.131	0.163	0.016	**0.011**	0.033	0.258
surv_colon	0.009	–	–	–	–	0.051	0.104	0.087	0.008	**0.003**	0.005	0.132
surv_grace	0.023	–	–	–	–	0.069	0.225	0.269	0.015	**0.005**	0.007	0.149
surv_trace	0.010	–	–	–	–	0.058	0.100	0.180	0.020	**0.005**	0.006	0.134
surv_framingham	0.025	–	–	–	–	0.060	0.297	0.226	0.035	0.009	**0.007**	0.205
tab_blood_transfusion	0.123	–	–	–	–	0.173	0.228	0.425	0.043	0.025	**0.024**	0.333
tab_credit_g	0.185	–	–	–	–	0.019	0.067	0.063	0.021	0.007	**0.004**	0.143
tab_qsar_biodeg	0.071	–	–	–	–	0.039	0.272	0.144	0.029	0.016	**0.008**	0.391

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
tab_maternal_risk	**0.043**	–	–	–	–	0.235	0.362	0.476	0.140	0.125	0.125	0.360
tab_website_phishing	0.100	–	–	–	–	0.005	0.060	0.003	0.008	**0.000**	0.001	0.028

Table 5: KS statistic (↓)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.190	0.201	0.335	**0.084**	0.262	0.162	0.356	0.246	0.101	0.092	0.089	0.313
breast_cancer	0.206	0.329	0.252	0.103	0.348	0.191	0.736	0.432	0.088	0.073	**0.059**	0.489
liver_disorders	0.229	0.415	0.265	0.074	0.073	0.241	0.278	0.358	0.123	**0.071**	0.080	0.479
parkinsons	0.184	0.382	0.268	0.098	0.293	0.253	0.301	0.454	0.112	0.106	**0.074**	0.454
obesity	0.258	0.239	0.235	0.216	0.156	0.217	0.354	0.289	0.161	0.123	**0.113**	0.305
diabetes	0.150	0.121	0.097	0.120	–	0.129	0.262	0.158	0.130	0.067	**0.046**	0.400
german_credit	0.118	0.125	0.381	0.046	–	0.099	0.249	0.143	0.095	0.043	**0.040**	0.359
bank_marketing	0.104	0.126	0.171	0.212	–	0.169	0.414	0.285	0.140	**0.061**	0.110	0.643
census_income	0.445	0.189	0.227	0.282	–	0.272	0.492	0.211	0.224	**0.126**	0.197	0.607
surv_melanoma	0.182	–	–	–	–	0.156	0.317	0.239	0.098	**0.068**	0.071	0.236
surv_pbc	0.116	–	–	–	–	0.119	0.189	0.147	0.103	0.032	**0.030**	0.174
surv_pbc3	0.143	–	–	–	–	0.178	0.347	0.302	0.103	0.080	**0.057**	0.360
surv_retinopathy	0.098	–	–	–	–	0.107	0.164	0.138	0.105	0.039	**0.032**	0.174
surv_whas500	0.115	–	–	–	–	0.092	0.230	0.185	0.065	0.040	**0.037**	0.268
surv_gbsg2	0.158	–	–	–	–	0.286	0.397	0.321	0.076	0.095	**0.054**	0.570
surv_colon	0.061	–	–	–	–	0.161	0.254	0.102	0.067	**0.032**	0.045	0.320
surv_grace	0.087	–	–	–	–	0.185	0.363	0.160	0.091	**0.030**	0.050	0.241
surv_trace	0.061	–	–	–	–	0.110	0.230	0.190	0.086	0.025	**0.024**	0.285
surv_framingham	0.079	–	–	–	–	0.132	0.436	0.297	0.122	0.042	**0.032**	0.316
tab_blood_transfusion	0.264	–	–	–	–	0.336	0.458	0.280	0.133	**0.067**	0.099	0.537
tab_credit_g	0.110	–	–	–	–	0.113	0.273	0.137	0.091	0.056	**0.042**	0.365
tab_qsar_biodeg	0.192	–	–	–	–	0.131	0.412	0.206	0.105	0.068	**0.061**	0.573
tab_maternal_risk	0.234	–	–	–	–	0.291	0.435	0.313	0.144	**0.111**	0.181	0.454

Table 6: KL divergence (↓)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.396	0.417	5.073	0.218	0.889	0.739	0.988	6.855	0.261	**0.189**	0.193	0.548
breast_cancer	0.499	0.531	1.804	0.260	1.136	0.522	6.607	11.996	0.173	0.146	**0.071**	0.801
liver_disorders	0.432	0.889	3.393	0.304	0.261	1.205	0.703	11.831	0.367	**0.191**	0.253	0.887
parkinsons	1.233	1.241	2.603	0.430	1.399	1.338	0.973	15.105	0.508	0.536	**0.396**	0.766
obesity	0.109	0.122	1.345	0.233	0.124	0.360	0.456	4.246	0.101	0.089	**0.026**	0.568
diabetes	0.104	0.079	0.360	0.175	–	0.380	0.584	2.691	0.093	0.088	**0.015**	0.570
german_credit	1.517	0.058	3.982	0.036	–	0.142	0.180	1.708	0.076	0.030	**0.015**	0.368
bank_marketing	0.180	**0.028**	0.615	0.111	–	0.235	0.300	2.197	0.133	0.039	0.039	0.819
census_income	0.450	**0.062**	0.987	0.155	–	0.463	1.749	3.233	0.257	0.090	0.132	1.137
surv_melanoma	0.355	–	–	–	–	0.812	1.628	8.892	0.200	0.198	**0.162**	0.343
surv_pbc	0.545	–	–	–	–	0.183	0.157	1.650	0.087	0.031	**0.011**	0.175
surv_pbc3	0.350	–	–	–	–	0.488	1.163	7.724	0.231	0.162	**0.092**	0.502
surv_retinopathy	2.306	–	–	–	–	0.106	0.178	2.314	0.059	0.016	**0.011**	0.092
surv_whas500	0.258	–	–	–	–	0.241	0.495	5.838	0.106	0.064	**0.048**	0.339
surv_gbsg2	0.311	–	–	–	–	0.586	0.609	4.183	0.138	**0.094**	0.153	0.780
surv_colon	0.092	–	–	–	–	0.273	0.616	2.297	0.031	**0.018**	0.023	0.350
surv_grace	0.289	–	–	–	–	0.862	2.834	6.324	0.085	**0.032**	0.034	0.384
surv_trace	0.093	–	–	–	–	0.304	0.298	3.033	0.059	**0.015**	0.017	0.331
surv_framingham	0.312	–	–	–	–	0.407	1.058	3.481	0.114	0.029	**0.021**	0.525
tab_blood_transfusion	1.269	–	–	–	–	2.008	1.265	11.252	0.251	0.167	**0.164**	0.906
tab_credit_g	3.869	–	–	–	–	0.098	0.366	1.565	0.064	0.023	**0.015**	0.377
tab_qsar_biodeg	0.776	–	–	–	–	0.296	3.983	3.813	0.180	0.109	**0.051**	1.125
tab_maternal_risk	0.715	–	–	–	–	1.545	3.575	12.353	0.412	**0.335**	0.452	1.023
tab_website_phishing	0.854	–	–	–	–	0.013	0.194	0.009	0.022	**0.001**	0.002	0.074

Table 7: Hellinger (↓)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.183	0.208	0.419	0.137	0.326	0.195	0.319	0.376	0.147	0.137	**0.134**	0.322
breast_cancer	0.209	0.313	0.297	0.138	0.490	0.219	0.668	0.636	0.132	0.111	**0.099**	0.477
liver_disorders	0.219	0.360	0.331	0.147	0.147	0.290	0.281	0.633	0.176	**0.144**	0.150	0.501
parkinsons	0.238	0.357	0.333	0.208	0.427	0.301	0.320	0.718	0.195	0.204	**0.178**	0.453
obesity	0.098	0.149	0.245	0.133	0.154	0.143	0.241	0.277	0.128	0.103	**0.059**	0.367
diabetes	0.118	0.135	0.142	0.148	–	0.161	0.302	0.208	0.133	0.116	**0.051**	0.402
german_credit	0.169	0.087	0.368	0.043	–	0.082	0.126	0.138	0.112	0.040	**0.039**	0.300
bank_marketing	0.102	0.069	0.163	0.065	–	0.112	0.200	0.178	0.140	0.066	**0.065**	0.416
census_income	0.182	**0.109**	0.186	0.111	–	0.194	0.308	0.274	0.213	0.124	0.126	0.491
surv_melanoma	0.189	–	–	–	–	0.200	0.328	0.436	0.144	0.120	**0.110**	0.269
surv_pbc	0.164	–	–	–	–	0.098	0.157	0.153	0.090	0.036	**0.035**	0.184
surv_pbc3	0.165	–	–	–	–	0.170	0.311	0.417	0.142	0.110	**0.093**	0.332
surv_retinopathy	0.212	–	–	–	–	0.105	0.140	0.144	0.094	0.042	**0.037**	0.115
surv_whas500	0.131	–	–	–	–	0.113	0.233	0.332	0.086	0.065	**0.060**	0.273
surv_gbsg2	0.162	–	–	–	–	0.206	0.254	0.294	0.100	**0.077**	0.116	0.396
surv_colon	0.070	–	–	–	–	0.145	0.187	0.162	0.062	**0.031**	0.045	0.262
surv_grace	0.119	–	–	–	–	0.193	0.341	0.383	0.094	**0.050**	0.062	0.271
surv_trace	0.069	–	–	–	–	0.148	0.218	0.275	0.100	**0.036**	0.040	0.282
surv_framingham	0.133	–	–	–	–	0.190	0.432	0.382	0.150	0.065	**0.058**	0.359
tab_blood_transfusion	0.289	–	–	–	–	0.364	0.399	0.609	0.178	0.138	**0.132**	0.499
tab_credit_g	0.287	–	–	–	–	0.086	0.165	0.135	0.099	0.041	**0.037**	0.304
tab_qsar_biodeg	0.208	–	–	–	–	0.144	0.401	0.290	0.133	0.095	**0.069**	0.550
tab_maternal_risk	**0.175**	–	–	–	–	0.397	0.513	0.618	0.304	0.288	0.287	0.538
tab_website_phishing	0.167	–	–	–	–	0.050	0.184	0.039	0.066	**0.015**	0.020	0.121

Table 8: Pearson corr. preservation (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.927	0.904	0.926	0.930	0.822	0.912	0.844	0.918	0.908	**0.939**	0.907	0.900
breast_cancer	0.868	0.606	0.799	**0.948**	0.773	0.887	0.566	0.753	0.694	0.899	0.945	0.607
liver_disorders	0.917	0.744	0.831	0.948	0.912	0.862	0.640	0.734	0.814	**0.949**	0.899	0.743
parkinsons	0.845	0.487	0.653	0.936	0.739	0.814	0.481	0.606	0.610	0.863	**0.950**	0.486
obesity	0.945	0.871	0.895	0.941	0.847	0.953	0.856	0.923	0.892	0.964	**0.974**	0.874
diabetes	0.928	0.898	0.934	0.916	–	0.968	0.882	0.905	0.918	**0.982**	0.956	0.868
german_credit	0.937	0.896	0.885	0.933	–	0.933	0.879	0.927	0.905	0.953	**0.961**	0.897
bank_marketing	0.952	0.943	0.896	0.966	–	0.969	0.922	0.969	0.947	**0.979**	0.965	0.940
census_income	0.951	0.953	0.942	0.980	–	0.970	0.914	0.961	0.946	0.980	**0.982**	0.941
surv_melanoma	0.910	–	–	–	–	0.855	0.794	0.848	0.865	**0.935**	0.924	0.827
surv_pbc	0.877	–	–	–	–	0.924	0.862	0.887	0.897	0.938	**0.943**	0.861
surv_pbc3	0.912	–	–	–	–	0.916	0.857	0.889	0.882	0.939	**0.942**	0.863
surv_retinopathy	0.958	–	–	–	–	0.868	0.784	0.852	0.813	0.899	**0.964**	0.779
surv_whas500	0.918	–	–	–	–	0.931	0.870	0.917	0.914	0.942	**0.944**	0.897
surv_gbsg2	0.902	–	–	–	–	0.882	0.803	0.884	0.887	**0.949**	0.921	0.848
surv_colon	0.953	–	–	–	–	0.915	0.911	0.947	0.931	**0.967**	0.940	0.917
surv_grace	0.951	–	–	–	–	0.942	0.883	0.918	0.950	**0.978**	0.923	0.919
surv_trace	0.965	–	–	–	–	0.931	0.853	0.877	0.924	0.977	**0.978**	0.876
surv_framingham	0.969	–	–	–	–	0.964	0.792	0.887	0.883	**0.980**	0.921	0.833
tab_blood_transfusion	0.683	–	–	–	–	0.880	0.576	0.680	0.731	**0.934**	0.864	0.549
tab_credit_g	0.959	–	–	–	–	0.930	0.889	0.930	0.903	0.954	**0.966**	0.888
tab_qsar_biodeg	0.791	–	–	–	–	0.906	0.809	0.873	0.837	0.933	**0.966**	0.829
tab_maternal_risk	0.916	–	–	–	–	0.906	0.751	0.882	0.822	**0.957**	0.910	0.750

Table 9: JS numeric (↓)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.054	0.094	0.200	0.042	0.194	0.070	0.192	0.265	0.040	0.041	**0.034**	0.189

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
breast_cancer	0.062	0.143	0.111	0.027	0.294	0.072	0.568	0.466	0.024	0.016	**0.013**	0.293
liver_disorders	0.068	0.184	0.129	0.028	0.028	0.113	0.118	0.450	0.040	**0.026**	0.029	0.316
parkinsons	0.082	0.175	0.136	0.056	0.227	0.120	0.143	0.578	0.049	0.053	**0.039**	0.269
obesity	0.040	0.081	0.136	0.114	0.072	0.093	0.237	0.313	0.052	0.059	**0.019**	0.206
diabetes	0.028	0.033	0.030	0.089	–	0.082	0.216	0.125	0.036	0.052	**0.009**	0.248
german_credit	0.032	0.047	0.202	0.023	–	0.044	0.111	0.181	0.031	0.019	**0.013**	0.160
bank_marketing	0.019	0.010	0.043	0.035	–	0.027	0.098	0.123	0.025	0.008	**0.004**	0.454
census_income	0.052	0.020	0.071	0.115	–	0.069	0.188	0.139	0.046	0.033	**0.019**	0.445
surv_melanoma	0.064	–	–	–	–	0.077	0.187	0.331	0.033	0.027	**0.023**	0.127
surv_pbc	0.039	–	–	–	–	0.025	0.062	0.111	0.016	0.005	**0.005**	0.058
surv_pbc3	0.042	–	–	–	–	0.055	0.197	0.330	0.030	0.024	**0.017**	0.194
surv_retinopathy	0.018	–	–	–	–	0.036	0.065	0.159	0.031	0.009	**0.007**	0.054
surv_whas500	0.034	–	–	–	–	0.030	0.129	0.244	0.015	0.010	**0.008**	0.129
surv_gbsg2	0.038	–	–	–	–	0.109	0.198	0.260	0.022	0.017	**0.014**	0.392
surv_colon	0.010	–	–	–	–	0.056	0.128	0.110	0.009	**0.004**	0.006	0.145
surv_grace	0.023	–	–	–	–	0.069	0.225	0.269	0.015	**0.005**	0.007	0.149
surv_trace	0.010	–	–	–	–	0.058	0.100	0.180	0.020	**0.005**	0.006	0.134
surv_framingham	0.025	–	–	–	–	0.060	0.297	0.226	0.035	0.009	**0.007**	0.205
tab_blood_transfusion	0.123	–	–	–	–	0.173	0.228	0.425	0.043	0.025	**0.024**	0.333
tab_credit_g	0.026	–	–	–	–	0.043	0.145	0.170	0.029	0.019	**0.010**	0.166
tab_qsar_biodeg	0.071	–	–	–	–	0.039	0.272	0.144	0.029	0.016	**0.008**	0.391
tab_maternal_risk	**0.043**	–	–	–	–	0.235	0.362	0.476	0.140	0.125	0.125	0.360

Table 10: JS categorical (↓)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
obesity	0.004	0.006	0.044	**0.000**	0.015	0.002	0.015	0.002	0.014	0.001	0.000	0.195
diabetes	0.010	0.020	0.049	**0.000**	–	0.003	0.047	0.002	0.022	0.000	0.000	0.155
german_credit	0.094	0.005	0.153	0.001	–	0.005	0.011	0.006	0.019	**0.001**	0.001	0.129
bank_marketing	0.014	0.008	0.038	**0.000**	–	0.050	0.079	0.052	0.051	0.018	0.023	0.135
census_income	0.049	0.018	0.043	**0.000**	–	0.092	0.177	0.150	0.114	0.040	0.076	0.274
surv_pbc	0.129	–	–	–	–	0.004	0.012	0.007	0.005	**0.000**	0.001	0.129
surv_pbc3	0.077	–	–	–	–	0.008	0.023	0.008	0.050	0.005	**0.003**	0.007
surv_retinopathy	0.270	–	–	–	–	0.011	0.013	0.001	0.006	0.001	**0.001**	0.003
surv_gbsg2	0.061	–	–	–	–	0.033	0.018	0.001	0.005	**0.000**	0.066	0.035
surv_colon	0.007	–	–	–	–	0.032	0.018	0.006	0.007	**0.001**	0.002	0.087
tab_credit_g	0.271	–	–	–	–	0.006	0.025	0.005	0.017	**0.001**	0.001	0.131
tab_website_phishing	0.100	–	–	–	–	0.005	0.060	0.003	0.008	**0.000**	0.001	0.028

Privacy / disclosure

Table 11: DCR median (↑ = less memorization)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.731	0.869	0.553	0.736	**1.971**	0.703	1.043	0.617	0.715	0.737	0.722	1.873
breast_cancer	0.550	1.245	0.476	0.555	**2.769**	0.672	1.726	0.433	0.850	0.607	0.488	2.494
liver_disorders	0.412	0.682	0.240	0.382	0.307	0.316	0.570	0.379	0.376	0.355	0.359	**1.677**
parkinsons	0.483	1.033	0.510	0.538	0.928	0.625	1.019	0.457	0.742	0.570	0.437	**1.923**
obesity	0.277	0.437	0.247	0.403	0.325	0.342	0.471	0.349	0.426	0.305	0.269	**0.706**
diabetes	0.053	0.082	0.038	0.116	–	0.065	0.118	0.068	0.081	0.079	0.060	**0.545**
german_credit	0.407	0.456	0.308	0.426	–	0.418	0.431	0.402	0.446	0.415	0.399	**0.642**
bank_marketing	0.160	0.254	0.096	0.269	–	0.250	0.331	0.226	0.289	0.247	0.243	**2.937**
census_income	0.087	0.195	0.044	0.254	–	0.188	0.247	0.189	0.222	0.195	0.196	**0.683**
surv_melanoma	0.364	–	–	–	–	0.375	0.748	0.348	0.419	0.376	0.379	**0.891**
surv_pbc	0.247	–	–	–	–	0.075	0.158	0.242	0.084	0.092	0.087	**0.571**
surv_pbc3	0.568	–	–	–	–	0.600	0.809	0.464	0.575	0.606	0.598	**1.414**
surv_retinopathy	0.342	–	–	–	–	0.300	0.454	0.321	0.373	0.285	0.230	**0.542**
surv_whas500	0.643	–	–	–	–	0.678	1.125	0.571	0.702	0.701	0.692	**1.822**

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
surv_gbsg2	0.311	–	–	–	–	0.324	0.534	0.261	0.334	0.306	0.371	**1.615**
surv_colon	0.196	–	–	–	–	0.271	0.430	0.142	0.206	0.215	0.202	**1.057**
surv_grace	0.228	–	–	–	–	0.207	0.676	0.191	0.209	0.231	0.236	**0.975**
surv_trace	0.081	–	–	–	–	0.072	0.230	0.238	0.085	0.083	0.079	**0.768**
surv_framingham	0.306	–	–	–	–	0.330	0.724	0.466	0.342	0.324	0.329	**1.356**
tab_blood_transfusion	0.260	–	–	–	–	0.172	0.313	0.244	0.230	0.196	0.160	**2.031**
tab_credit_g	0.422	–	–	–	–	0.405	0.453	0.395	0.432	0.416	0.395	**0.642**
tab_qsar_biodeg	0.474	–	–	–	–	0.591	1.033	0.482	0.829	0.473	0.376	**4.355**
tab_maternal_risk	0.260	–	–	–	–	0.353	0.486	0.307	0.440	0.359	0.373	**1.381**
tab_website_phishing	0.322	–	–	–	–	0.283	**0.400**	0.283	0.322	0.283	0.283	**0.400**

Table 12: DCR real–real reference

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.801	0.801	0.801	0.801	0.796	0.801	0.801	0.801	0.801	0.801	0.801	0.801
breast_cancer	0.465	0.465	0.465	0.465	0.468	0.465	0.465	0.465	0.465	0.465	0.465	0.465
liver_disorders	0.380	0.380	0.380	0.380	0.342	0.380	0.380	0.380	0.380	0.380	0.380	0.380
parkinsons	0.524	0.524	0.524	0.524	0.538	0.524	0.524	0.524	0.524	0.524	0.524	0.524
obesity	0.279	0.279	0.279	0.279	0.284	0.276	0.279	0.279	0.279	0.279	0.279	0.279
diabetes	0.071	0.071	0.071	0.071	–	0.071	0.071	0.071	0.071	0.071	0.071	0.071
german_credit	0.423	0.419	0.419	0.419	–	0.419	0.419	0.419	0.419	0.419	0.419	0.419
bank_marketing	0.209	0.209	0.209	0.209	–	0.209	0.209	0.209	0.209	0.209	0.209	0.209
census_income	0.136	0.136	0.136	0.136	–	0.136	0.136	0.136	0.136	0.136	0.136	0.136
surv_melanoma	0.524	–	–	–	–	0.524	0.524	0.524	0.524	0.524	0.524	0.524
surv_pbc	0.196	–	–	–	–	0.196	0.196	0.196	0.196	0.196	0.196	0.196
surv_pbc3	0.626	–	–	–	–	0.626	0.626	0.626	0.626	0.626	0.626	0.626
surv_retinopathy	0.318	–	–	–	–	0.318	0.318	0.318	0.318	0.318	0.318	0.318
surv_whas500	0.745	–	–	–	–	0.745	0.745	0.745	0.745	0.745	0.745	0.745
surv_gbsg2	0.323	–	–	–	–	0.323	0.323	0.323	0.323	0.323	0.323	0.323
surv_colon	0.313	–	–	–	–	0.337	0.337	0.337	0.337	0.337	0.337	0.337
surv_grace	0.290	–	–	–	–	0.290	0.290	0.290	0.290	0.290	0.290	0.290
surv_trace	0.109	–	–	–	–	0.109	0.109	0.109	0.109	0.109	0.109	0.109
surv_framingham	0.352	–	–	–	–	0.352	0.352	0.352	0.352	0.352	0.352	0.352
tab_blood_transfusion	0.201	–	–	–	–	0.201	0.201	0.201	0.201	0.201	0.201	0.201
tab_credit_g	0.410	–	–	–	–	0.410	0.410	0.410	0.410	0.410	0.410	0.410
tab_qsar_biodeg	0.436	–	–	–	–	0.436	0.436	0.436	0.436	0.436	0.436	0.436
tab_maternal_risk	0.357	–	–	–	–	0.357	0.357	0.357	0.357	0.357	0.357	0.357
tab_website_phishing	0.400	–	–	–	–	0.400	0.400	0.400	0.400	0.400	0.400	0.400

Table 13: DCR (encoded) (↑)

Dataset	DISCO-TAB
heart_failure	**2.153**
breast_cancer	**3.280**
liver_disorders	**0.641**
parkinsons	**2.372**
obesity	**1.166**
diabetes	**0.498**
german_credit	**2.974**
bank_marketing	**1.652**
census_income	**1.640**

Table 14: NNDR (↑)

Dataset	DISCO-TAB
heart_failure	**0.892**
breast_cancer	**0.980**
liver_disorders	**0.922**
parkinsons	**0.951**
obesity	**0.902**
diabetes	**0.831**
german_credit	**0.953**
bank_marketing	**0.927**
census_income	**0.931**

Table 15: NNAA (0.5 ideal)

Dataset	DISCO-TAB
heart_failure	0.604
breast_cancer	0.975
liver_disorders	0.361
parkinsons	0.903
obesity	0.591
diabetes	0.585
german_credit	0.790
bank_marketing	0.612
census_income	0.663

Table 16: MIA AUROC (0.5 safe)

Dataset	DISCO-TAB
heart_failure	0.499
breast_cancer	0.516
liver_disorders	0.632
parkinsons	0.516
obesity	0.559
diabetes	0.504
german_credit	0.513
bank_marketing	0.505
census_income	0.503

Table 17: Duplicate rate (↓)

Dataset	DISCO-TAB
heart_failure	**0.000**
breast_cancer	**0.000**
liver_disorders	**0.307**
parkinsons	**0.000**
obesity	**0.000**
diabetes	**0.000**
german_credit	**0.000**
bank_marketing	**0.000**
census_income	**0.000**

Diversity / coverage (PRDC)

Table 18: Precision (↑)

Dataset	DISCO-TAB
heart_failure	**0.965**
breast_cancer	**0.575**
liver_disorders	**0.995**
parkinsons	**0.872**
obesity	**0.937**
diabetes	**0.982**
german_credit	**0.924**
bank_marketing	**0.971**
census_income	**0.967**

Table 19: Recall (↑)

Dataset	DISCO-TAB
heart_failure	**0.921**
breast_cancer	**0.103**
liver_disorders	**0.918**
parkinsons	**0.372**
obesity	**0.923**
diabetes	**0.945**
german_credit	**0.473**
bank_marketing	**0.790**
census_income	**0.712**

Table 20: Density (↑)

Dataset	DISCO-TAB
heart_failure	**0.866**
breast_cancer	**0.311**
liver_disorders	**1.139**
parkinsons	**0.960**
obesity	**0.904**
diabetes	**1.021**
german_credit	**0.742**
bank_marketing	**1.104**
census_income	**1.189**

Table 21: Coverage (↑)

Dataset	DISCO-TAB
heart_failure	**0.841**
breast_cancer	**0.217**
liver_disorders	**0.899**
parkinsons	**0.556**
obesity	**0.882**
diabetes	**0.907**
german_credit	**0.783**
bank_marketing	**0.898**
census_income	**0.841**

Detection / distribution

Table 22: C2ST AUROC (↓, 0.5 best)

Dataset	DISCO-TAB
heart_failure	**0.969**
breast_cancer	**1.000**
liver_disorders	**0.662**
parkinsons	**0.999**
obesity	**0.998**
diabetes	**1.000**
german_credit	**1.000**
bank_marketing	**0.888**
census_income	**1.000**

Table 23: MMD^2 (↓)

Dataset	DISCO-TAB
heart_failure	**0.050**
breast_cancer	**0.104**
liver_disorders	**0.051**
parkinsons	**0.083**
obesity	**0.005**
diabetes	**0.024**
german_credit	**0.046**
bank_marketing	**0.012**
census_income	**0.039**

Table 24: Fréchet-PCA dist. (↓)

Dataset	DISCO-TAB
heart_failure	**2.197**
breast_cancer	**24.230**
liver_disorders	**0.799**
parkinsons	**6.899**
obesity	**0.287**
diabetes	**1.776**
german_credit	**1.858**
bank_marketing	**0.856**
census_income	**1.344**

Table 25: Wasserstein (↓)

Dataset	DISCO-TAB
heart_failure	**0.319**
breast_cancer	**0.309**
liver_disorders	**0.378**
parkinsons	**0.264**
obesity	**0.119**
diabetes	**0.235**
german_credit	**0.225**
bank_marketing	**0.162**
census_income	**0.279**

Table 26: Association error (↓)

Dataset	DISCO-TAB
heart_failure	**0.057**
breast_cancer	**0.278**
liver_disorders	**0.077**
parkinsons	**0.242**
obesity	**0.047**
diabetes	**0.055**
german_credit	**0.108**
bank_marketing	**0.038**
census_income	**0.059**

SDMetrics quality report

Table 27: Overall quality (↑)

Dataset	DISCO-TAB
heart_failure	**0.374**
breast_cancer	**0.707**
liver_disorders	**0.805**
parkinsons	**0.766**
obesity	**0.843**
diabetes	**0.385**
german_credit	**0.715**
bank_marketing	**0.704**
census_income	**0.518**

Table 28: Column shapes (↑)

Dataset	DISCO-TAB
heart_failure	**0.748**
breast_cancer	**0.756**
liver_disorders	**0.643**
parkinsons	**0.773**
obesity	**0.851**
diabetes	**0.770**
german_credit	**0.799**
bank_marketing	**0.856**
census_income	**0.663**

Table 29: Column-pair trends (↑)

Dataset	DISCO-TAB
heart_failure	**0.000**
breast_cancer	**0.658**
liver_disorders	**0.967**
parkinsons	**0.759**
obesity	**0.834**
diabetes	**0.000**
german_credit	**0.630**
bank_marketing	**0.552**
census_income	**0.373**

FAITH framework

Table 30: FAITH overall (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.813	0.884	0.739	0.827	0.870	0.808	0.916	0.758	0.804	0.834	0.817	**0.967**
breast_cancer	0.388	0.883	0.804	0.897	0.856	**0.925**	0.870	0.770	0.900	0.899	0.854	0.883
liver_disorders	0.870	0.903	0.710	0.859	0.835	0.800	0.845	0.804	0.819	0.852	0.837	**0.923**
parkinsons	0.298	0.852	0.763	0.868	0.825	**0.877**	0.847	0.715	0.860	0.862	0.818	0.852
obesity	0.561	0.944	0.790	0.957	0.854	0.902	0.941	0.907	0.947	0.877	0.855	**0.961**
diabetes	0.745	0.862	0.769	0.924	–	0.853	0.877	0.841	0.865	0.886	0.842	**0.964**
german_credit	0.333	0.881	0.724	0.859	–	0.847	0.853	0.814	0.878	0.852	0.829	**0.960**
bank_marketing	0.810	0.922	0.749	0.936	–	0.932	0.960	0.903	0.941	0.927	0.919	**0.974**
census_income	0.350	0.949	0.769	0.982	–	0.957	0.974	0.967	0.976	0.959	0.935	**0.984**
surv_melanoma	0.791	–	–	–	–	0.772	0.880	0.763	0.804	0.806	0.808	**0.902**
surv_pbc	0.655	–	–	–	–	0.797	0.828	0.836	0.797	0.812	0.812	**0.913**
surv_pbc3	0.795	–	–	–	–	0.827	0.907	0.742	0.808	0.839	0.833	**0.959**
surv_retinopathy	0.657	–	–	–	–	0.807	0.830	0.818	0.817	0.809	0.804	**0.859**
surv_whas500	0.295	–	–	–	–	0.822	0.942	0.761	0.830	0.838	0.839	**0.965**
surv_gbsg2	0.664	–	–	–	–	0.840	0.887	0.802	0.843	0.853	0.869	**0.954**
surv_colon	0.822	–	–	–	–	0.845	0.887	0.805	0.822	0.841	0.830	**0.956**
surv_grace	0.803	–	–	–	–	0.792	0.915	0.767	0.795	0.817	0.805	**0.953**
surv_trace	0.804	–	–	–	–	0.801	0.878	0.867	0.820	0.827	0.826	**0.944**
surv_framingham	0.816	–	–	–	–	0.839	0.916	0.873	0.823	0.839	0.826	**0.939**
tab_blood_transfusion	0.474	–	–	–	–	0.798	0.836	0.811	0.811	0.834	0.787	**0.881**
tab_credit_g	0.379	–	–	–	–	0.840	0.888	0.826	0.872	0.866	0.841	**0.958**
tab_qsar_biodeg	0.363	–	–	–	–	0.921	0.938	0.865	0.938	0.883	0.847	**0.938**
tab_maternal_risk	0.527	–	–	–	–	0.834	0.872	0.789	0.847	0.853	0.844	**0.925**
tab_website_phishing	0.351	–	–	–	–	0.509	0.530	0.509	0.514	0.507	0.509	**0.581**

Table 31: Factuality (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.978	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
breast_cancer	0.000	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
liver_disorders	0.999	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
parkinsons	0.000	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
obesity	0.416	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
diabetes	0.859	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
german_credit	0.000	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
bank_marketing	**1.000**	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
census_income	0.091	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_melanoma	0.998	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_pbc	0.621	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_pbc3	0.986	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_retinopathy	0.544	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_whas500	0.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_gbsg2	0.631	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_colon	**1.000**	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_grace	**1.000**	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_trace	1.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_framingham	1.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_blood_transfusion	0.313	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_credit_g	0.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_qsar_biodeg	0.029	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_maternal_risk	0.453	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_website_phishing	0.667	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Table 32: Alignment (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.910	0.877	**0.944**	0.916	0.790	0.893	0.808	0.904	0.887	0.927	0.887	0.879
breast_cancer	0.882	0.532	0.747	0.933	0.722	0.861	0.479	0.713	0.627	0.873	**0.934**	0.532
liver_disorders	0.917	0.707	0.789	**0.943**	0.901	0.858	0.600	0.725	0.782	0.941	0.896	0.693
parkinsons	0.832	0.411	0.598	0.919	0.678	0.767	0.393	0.545	0.547	0.844	**0.933**	0.409
obesity	0.936	0.845	0.855	0.929	0.824	0.946	0.819	0.903	0.866	0.958	**0.971**	0.845
diabetes	0.912	0.888	0.915	0.910	–	0.965	0.868	0.903	0.906	**0.979**	0.946	0.859
german_credit	0.942	0.841	0.892	0.914	–	0.919	0.858	0.901	0.863	0.943	**0.956**	0.843
bank_marketing	0.932	0.936	0.877	0.919	–	0.960	0.891	0.959	0.912	**0.970**	0.959	0.897
census_income	0.938	0.941	0.929	0.974	–	0.964	0.905	0.956	0.939	0.978	**0.979**	0.935
surv_melanoma	0.903	–	–	–	–	0.841	0.772	0.831	0.853	**0.931**	0.911	0.811
surv_pbc	0.879	–	–	–	–	0.913	0.853	0.877	0.891	0.935	**0.940**	0.856
surv_pbc3	0.892	–	–	–	–	0.898	0.816	0.863	0.859	0.925	**0.929**	0.836
surv_retinopathy	0.953	–	–	–	–	0.830	0.642	0.775	0.693	0.890	**0.957**	0.630
surv_whas500	0.897	–	–	–	–	0.917	0.824	0.900	0.892	0.928	**0.933**	0.865
surv_gbsg2	0.898	–	–	–	–	0.865	0.777	0.862	0.865	**0.944**	0.903	0.818
surv_colon	0.947	–	–	–	–	0.901	0.840	0.927	0.900	**0.960**	0.932	0.842
surv_grace	0.946	–	–	–	–	0.937	0.855	0.914	0.947	**0.975**	0.913	0.903
surv_trace	0.958	–	–	–	–	0.922	0.824	0.859	0.913	0.974	**0.975**	0.852
surv_framingham	0.964	–	–	–	–	0.957	0.724	0.849	0.842	**0.974**	0.898	0.766
tab_blood_transfusion	0.718	–	–	–	–	0.867	0.564	0.649	0.718	**0.926**	0.831	0.530
tab_credit_g	0.953	–	–	–	–	0.920	0.853	0.903	0.865	0.944	**0.963**	0.831
tab_qsar_biodeg	0.732	–	–	–	–	0.864	0.751	0.820	0.768	0.903	**0.952**	0.754
tab_maternal_risk	0.908	–	–	–	–	0.896	0.699	0.853	0.789	**0.948**	0.897	0.713

Table 33: Integrity (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.386	0.660	0.013	0.393	0.690	0.338	0.856	0.126	0.329	0.407	0.381	**0.990**
breast_cancer	0.670	**1.000**	0.470	0.656	0.703	0.838	**1.000**	0.367	0.971	0.724	0.482	**1.000**
liver_disorders	0.565	0.906	0.051	0.494	0.438	0.342	0.781	0.493	0.494	0.465	0.450	**0.998**
parkinsons	0.361	0.998	0.453	0.553	0.622	0.739	0.994	0.314	0.895	0.605	0.338	**1.000**
obesity	0.478	0.931	0.304	0.900	0.590	0.663	0.943	0.725	0.921	0.549	0.451	**1.000**
diabetes	0.350	0.558	0.160	0.784	–	0.446	0.638	0.459	0.555	0.566	0.420	**0.996**
german_credit	0.388	0.685	0.006	0.521	–	0.470	0.553	0.356	0.647	0.464	0.361	**0.998**
bank_marketing	0.309	0.754	0.118	0.823	–	0.769	0.951	0.654	0.853	0.737	0.716	**1.000**
census_income	0.280	0.855	0.147	0.955	–	0.863	0.991	0.913	0.964	0.858	0.762	**1.000**
surv_melanoma	0.264	–	–	–	–	0.246	0.750	0.220	0.364	0.295	0.321	**0.799**
surv_pbc	0.498	–	–	–	–	0.276	0.458	0.469	0.299	0.312	0.309	**0.795**
surv_pbc3	0.315	–	–	–	–	0.410	0.811	0.103	0.372	0.429	0.401	**0.999**
surv_retinopathy	0.585	–	–	–	–	0.398	0.677	0.495	0.576	0.347	0.261	**0.805**
surv_whas500	0.283	–	–	–	–	0.371	0.942	0.145	0.426	0.426	0.424	**0.995**
surv_gbsg2	0.495	–	–	–	–	0.495	0.769	0.345	0.505	0.467	0.574	**0.998**
surv_colon	0.342	–	–	–	–	0.477	0.706	0.293	0.387	0.405	0.388	**0.983**
surv_grace	0.265	–	–	–	–	0.233	0.805	0.154	0.233	0.294	0.308	**0.909**
surv_trace	0.259	–	–	–	–	0.280	0.687	0.610	0.366	0.336	0.329	**0.924**
surv_framingham	0.301	–	–	–	–	0.399	0.941	0.643	0.450	0.383	0.405	**0.991**
tab_blood_transfusion	0.551	–	–	–	–	0.326	0.782	0.596	0.528	0.410	0.316	**0.995**
tab_credit_g	0.561	–	–	–	–	0.438	0.700	0.399	0.624	0.520	0.400	**0.999**
tab_qsar_biodeg	0.663	–	–	–	–	0.821	**1.000**	0.638	0.984	0.629	0.437	**1.000**
tab_maternal_risk	0.295	–	–	–	–	0.442	0.789	0.302	0.599	0.463	0.481	**0.989**
tab_website_phishing	0.069	–	–	–	–	0.035	0.121	0.038	0.058	0.029	0.035	**0.325**

Table 34: Tracking (↑)

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.978	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
breast_cancer	0.000	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
liver_disorders	0.999	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
parkinsons	0.000	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
obesity	0.416	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
diabetes	0.859	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
german_credit	0.000	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
bank_marketing	**1.000**	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
census_income	0.091	**1.000**	**1.000**	**1.000**	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_melanoma	0.998	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_pbc	0.621	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_pbc3	0.986	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_retinopathy	0.544	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_whas500	0.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_gbsg2	0.631	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_colon	**1.000**	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_grace	**1.000**	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_trace	1.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
surv_framingham	1.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_blood_transfusion	0.313	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_credit_g	0.000	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_qsar_biodeg	0.029	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_maternal_risk	0.453	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
tab_website_phishing	0.667	–	–	–	–	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Other

Table 35: Parse valid-rate (↑)

Dataset	DISCO-TAB
heart_failure	**0.675**
breast_cancer	**0.693**
liver_disorders	**0.680**
parkinsons	**0.655**
obesity	**0.708**
diabetes	**0.711**
german_credit	**0.702**
bank_marketing	**0.711**
census_income	**0.711**
surv_melanoma	**0.659**
surv_pbc	**0.474**
surv_pbc3	**0.680**
surv_retinopathy	**0.682**
surv_whas500	**0.688**
surv_gbsg2	**0.696**
surv_colon	**0.700**
surv_grace	**0.702**
surv_trace	**0.708**
surv_framingham	**0.711**
tab_blood_transfusion	**0.529**
tab_credit_g	**0.702**
tab_qsar_biodeg	**0.703**
tab_maternal_risk	**0.688**
tab_website_phishing	**0.698**